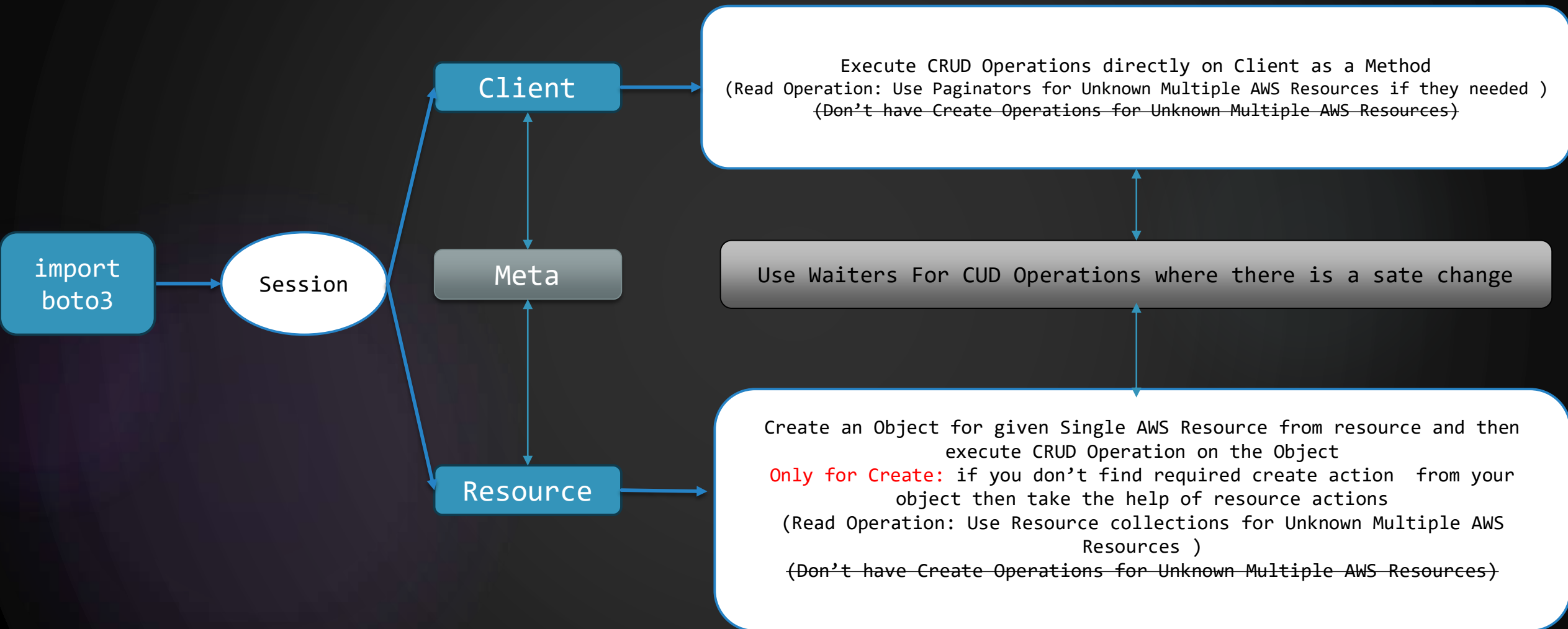


Introduction to Boto3 CRUD Operations for Unknown Multiple AWS Resources

- Multiple AWS Resources means like IAM Users, IAM Roles, IAM Policies, EC2 Instances, S3 Buckets...
- Notes for Unknown Multiple AWS Resources:
 - "Unknown" here does not mean you (the developer) don't know
 - It means the script doesn't know the exact AWS resources yet
 - (Because when you log into the AWS console (UI), you can see all resources manually)
 - Create Operations are not applicable to unknown Multiple AWS Resources because you cannot "create" something unknown
 - To execute Update and Delete Operations first we have to discover or find or read them
 - Once you discover or find or read unknown AWS Resources then you can execute Update and Delete Operations using our single AWS Resource concepts
 - Boto3 provides paginators, collections and filters to discover or find or read unknowns Multiple AWS Resources
 - Paginators for client operations
 - Collection for Resource Operations
 - Filters (narrow down the search results) (Generally for both client and resource but not every resource method supports filtering)
- Note: We may get Unknown Single AWS Resource result while working with Unknown Multiple AWS Resources

Steps to Develop Boto3 Scripts



Notes for Boto3 Scripts

- Single AWS Resource:
 - Client Operations:
 - Create Operations : create/put/allocate/add...
 - Read Operations : get/list/describe ...
 - Update Operations : update/modify ...
 - Delete Operations : delete/remove/terminate ...
 - Client CRUD Operations return results as a dictionary/JSON Data
 - So, we need dictionary operations to parse result
 - Resource Operations:
 - CRUD Operations are using single AWS Resource object
 - CUD Operations are executed with Actions on Single AWS Resource Object
 - Note: If you don't find action for create from single AWS Resource object then execute from Resource action
 - CUD Operations return results as a dictionary or list or list of objects
 - R Operations are executed with the help of Identifiers and Attributes
- Known and Unknown Multiple AWS Resources:
 - Known multiple AWS resource operations are performed using Single AWS Resource operations inside loops
 - Unknown multiple AWS resources are retrieved using paginators, collections or filters and update or delete operations are performed using Single AWS Resource operations inside loops
 - Note: We may get Unknown Single AWS Resource result while working with Unknown Multiple AWS Resources